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$$\int x\sin^2(x)dx$$
$$= \int x \cdot \frac{1-\cos 2x}{2}dx$$
$$= \frac{1}{2}\int xdx - \frac{1}{2}\int x\cos(2x)dx$$
$$\begin{cases} u = x \\ dv = \cos(2x)dx \end{cases} \Rightarrow \begin{cases} du = dx \\ v = \dfrac{\sin 2x}{2} \end{cases}$$
$$\int x\cos(2x)dx = \frac{x}{2}\sin 2x - \frac{1}{2}\int \sin 2xdx$$
$$= \frac{x}{2}\sin 2x + \frac{\cos 2x}{4}$$
$$\int x\sin^2 xdx = \frac{x^2}{4} - \frac{1}{2}\left(\frac{x}{2}\sin 2x + \frac{1}{4}\cos 2x\right) + C$$
$$= \frac{x^2}{4} - \frac{x}{4}\sin 2x - \frac{\cos 2x}{8} + C$$

References